TriAlign Technical Design

An Agent for Clinically Meaningful Comparison of Eligibility Criteria

Architecture, Dataset, Pair Semantics, Interface Concepts, and Research Notes

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Abstract

TriAlign compares clinical-trial eligibility rules using structured constraints, dense representations, and contextual reasoning. This document defines what constitutes a clinically meaningful criterion relationship, how source evidence and normalization are retained, how an agent selects and reconciles comparison tools, and how a clinician could inspect and correct a result. The frozen criterion release contains 3,537 records from 100 trials. Pair labels, trained models, routing evaluation, and interface concepts described here are subsequent deliverables, and in the future, experimental results will be added.

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1 Motivation

1.1 The Clinical Problem

An eligibility criterion determines what characteristics a patient must have to be included in a clinical trial. Two rules can reuse the same terminology while admitting different people, ie "age $\geq$ 18 years" vs "age $<$ 18 years" have similar vocabulary and disjoint allowed age ranges. Conversely, an abbreviation and an expanded term may express the same rule with limited word overlap. TriAlign is intended to pull seemingly similar criteria apart when clinical investigators, protocol designers, and research coordinators compare protocols.

The primary workflow is criterion-to-criterion comparison across trials: select a current rule, retrieve historical rules about the same clinical concept, inspect the operational differences, and record an evidence-supported relationship. Potential applications include identifying precedent during protocol design, comparing amendments, documenting harmonization decisions, and inspecting whether superficially similar study populations are defined by materially different criteria. These applications motivate the system; their clinical benefit must be measured rather than assumed.

TriAlign does not determine an individual patient's eligibility, recommend a treatment, estimate recruitment yield without patient data, or establish causal transportability. A criterion-level comparison is not a claim that two complete protocols recruit the same population: their remaining rules, sites, ascertainment practices, and treatment settings may differ.

1.2 Why textual similarity is insufficient

The clinical meaning of a rule depends on more than its entity name.

Dimension	Illustrative difference	Comparison implication
Polarity	Require a condition versus exclude that condition.	Interpret the allowed population before using numerical or categorical agreement.
Threshold and boundary	Age ≥ 18 versus > 18.	A one-point boundary changes equivalence and containment.
Units and references	Absolute bilirubin concentration versus a multiple of ULN.	Equal numbers need not be comparable quantities.
Time and anchor	Event in the previous six months versus six months after treatment.	Duration, direction, and reference event must agree.
Repetition	A single observation versus two observations on separate occasions.	An identical threshold can encode a stronger evidentiary requirement.
Disease specificity	Active disease, previous disease, anatomical site, stage, or severity.	Ontological similarity alone does not establish equal eligibility.
Exceptions	Other malignancy excluded except specified cured cancers.	Removing the exception reverses the interpretation of an allowed subgroup.
Boolean scope	$A \wedge B$ versus $A \vee B$, possibly inside a larger clause.	Independent atoms must retain their branch conditions.

An embedding score can be useful for finding a likely comparison without being a valid clinical relationship score. In particular, cosine similarity is symmetric, whereas "A is more restrictive than B" is directional. A high cosine score therefore cannot, by itself, select a directional label or prove equivalence.

1.3 Research questions and hypotheses

Three questions we.re interested in are:

1. Which phenomena produce false-high similarity for clinically different rules, or false-low similarity for equivalent rules, in general and biomedical encoders?

2. Does supervision from reviewed positives, ordinary negatives, hard negatives, and ordered anchors improve held-out retrieval and relationship prediction?

3. Does selective tool routing improve evidence support, calibration, reviewer effort, latency, or cost relative to an LLM-only judge and an always-run fixed chain?

2 Eligibility Criteria Dataset

2.1 Modern AutoCrit and the upstream handoff

Modern AutoCrit produces the source records used by TriAlign. It accepts an NCT identifier, protocol text, or a protocol PDF; locates eligibility content; distinguishes inclusion from exclusion; extracts atomic constraints; and preserves the parent statement and local context. Extraction records include clinical fields, source identifiers, protocol versions, validation findings, and review status. TriAlign consumes reviewed outputs rather than replacing this extraction system. Criterion review corrects segmentation, polarity, measurement fields, normalization, and scope.

2.2 Current frozen release and historical counts

Release property	Verified value or source
Dataset version	`trial100_v1`
Source workbook	`data/trial100_v1.xlsx`, sheet `Human Review`
Frozen snapshot	`data/processed/trial100_v1.parquet`
Manifest	`data/processed/trial100_v1.parquet.manifest.json`
Retained criteria / trials	3,537 / 100
Review status	3,537 corrected; zero pending
Export readiness	All retained rows export-ready; zero recorded validation issues or readiness blockers
Schema version	`0.1.0`
Pair labels	Not included in the criterion release

Table 2: Current release facts are drawn from the frozen manifest. A review-status flag is distinct from a pair label or a calibrated confidence estimate.

The cohort design in the proposal is 50 oncology and 50 cardiovascular trials. Therapeutic-area evaluation will use an explicit trial-level mapping; that design statement is not a verified per-row therapeutic-area annotation in the manifest.

The initial extraction had 3,145 candidates. Splitting, correction, and removal changed the number of retained records. The October 20 proposal milestone reported approximately 2,900 reviewed cases and a remaining uncertainty queue. Those historical workload figures are not the denominators for v1 evaluation. Use the frozen manifest and saved split files for every reported result.

2.3 Record schema and evidence hierarchy

Field group	Representative fields	Technical use
Identity and lineage	`criterion_id`, `row_id`, `trial_id`, `parent_statement_id`, `origin_criterion_id`	Stable joins, split membership, split-parent lineage, and audit history.

Field group	Representative fields	Technical use
Source evidence	`source`, `source_excerpt`, `parent_statement`, `surrounding_context`	Human inspection, evidence citations, and scope reconstruction.
Clinical predicate	`criterion_type`, `domain`, `entity`, `comparator`, `value`, `unit`, `value_kind`, bounds and inclusivity	Reconstruct the stated constraint before interpreting its eligibility effect.
Context and logic	`negated`, `temporal`, `repetition`, `qualifier`, `logical_operator`, `boolean_group_id`, atom indices/counts	Retain conditions and connections among atoms; flat fields alone do not certify a complete Boolean tree.
Normalization	Raw and normalized entity/value/unit fields; normalized bounds; `reference_type`, `reference_multiplier`; methods/status/confidence	Controlled matching while preserving source-grounded values and normalization provenance.
Protocol provenance	`source_uri`, `source_version`, `source_retrieved_at`, `protocol_hash`, `source_id`	Identify the exact source snapshot rather than silently following a changed registry page.
Review and release	`review_status`, `export_ready`, issue/blocker lists, review-method metadata	Admission to the release and traceability of subsequent corrections.

The evidence hierarchy is the protocol wording and its governing context, followed by the reviewed structured interpretation and normalization. When an atom and its parent appear inconsistent, the system should expose the conflict and request adjudication. Opening a current registry link does not establish that its wording matches the version originally extracted.

2.4 Normalization and serialization

Terminology normalization supports retrieval across abbreviations, spelling variants, and aliases. It must preserve disease stage, severity, anatomical site, assay, and treatment context. A normalization dictionary should carry its own version and provenance, and proposed synonym merges should be evaluated for both false merges and missed aliases. A method/status field can describe normalization without establishing that its numeric confidence is calibrated.

Physical-unit normalization and reference normalization are different. "Bilirubin $\leq$ 1.5 $\times$ ULN" uses `reference_type=ULN` and multiplier 1.5. ULN is not a physical concentration unit. A local upper limit can vary by assay or institution, so reference-relative and absolute thresholds require explicit reference information before conversion. Likewise, systolic/diastolic notation does not authorize inferring an AND/OR relationship that the source did not specify.

The source's clinical and normalization fields are retained in the snapshot. Schema loading standardizes missing values, Boolean fields, issue lists, and the casing of selected categorical status fields. Because Parquet requires homogeneous column types, the mixed `value`, `value_raw`, and `value_normalized` columns use manifest-declared JSON scalar encoding. The project loader decodes these columns, preserving numeric values, categorical strings such as "001", and Booleans. Numerical bounds remain typed numeric columns. Direct consumers of Parquet must read the manifest; they must not interpret JSON scalar text as already-decoded clinical values.

2.5 Release integrity and changes after freezing

The release records the source and Parquet hashes, schema version, criterion-ID fingerprint, creation time, Git revision, and exporter/script hashes. The export was reopened through the project loader and reconciled record

by record against source-derived records. These checks establish serialization integrity, not independent clinical certification of every interpretation.

A correction after freezing creates a new dataset version, an explicit change record, and a regenerated manifest. Existing pair records must continue to identify the corpus version against which they were labeled. A source edit must not silently change an already-published training or test example. Shared public-source subsets should retain this provenance and an intended-use statement. Sharing the criterion corpus with the group mentioned in the October presentation feedback must not expose protected TriAlign test labels.

2.6 Optional external data

Chia is an annotated eligibility-criteria corpus with entity, relation, and scope representations [14]. It may supply external terminology or logical-complexity examples. Its annotations do not directly provide TriAlign's eight pair relationships. Any use requires an explicit schema mapping, license/source review, overlap checks, and separate reporting; it must not be silently mixed into the v1 benchmark.

3 Clinical Pair Definitions

3.1 The ordered comparison unit

A pair contains two reviewed criteria, A and B, normally from different protocols, together with their corpus versions, trial/criterion identifiers, source excerpts, parent context, structured fields, and sampling provenance. The input order is part of the task. An unordered pair-group identifier can deduplicate cases and hold reversed views together, while an ordered pair identifier or orientation field preserves the direction of prediction. Mirrored cases must not become independent examples across train and test.

Criterion approval establishes that one record represents its source rule. It does not establish a relationship to any other criterion. Pair labeling is a second review with its own evidence, uncertainty, and adjudication record. The following formalization is an operational draft of the labeling contract; JM must approve the final guide before benchmark labels are frozen.

3.2 Allowed populations and comparability

Let U_γ be a common comparison universe under explicit context γ, including the measurement, population, temporal anchor, and branch scope. Let $P_c(x;\gamma)$ be the clinical predicate stated by criterion c, with its internal negation and exceptions already interpreted. The criterion-level allowed set is

$$S(c;\gamma) = \begin{cases} \{x \in U_\gamma : P_c(x;\gamma)\}, & c \text{ is an inclusion rule,} \\ U_\gamma \setminus \{x : P_c(x;\gamma)\}, & c \text{ is an exclusion rule.} \end{cases}$$

An internal negation is applied inside P_c; inclusion/exclusion is applied once outside it. A categorical value "absent" and a negation flag must not accidentally invert a predicate twice. Record their combined interpretation before comparing sets.

Set operations are meaningful only after establishing a common operational frame. Age and platelet count can both restrict a person, but this does not make their comparison a useful "partial overlap" label. First determine conceptual comparability. Related measurements such as serum creatinine and estimated glomerular filtration rate may require a clinical mapping that is not available; terminology similarity alone is insufficient.

3.3 Eight relationship labels

For comparable, interpretable criteria, write $S_A = S(A;\gamma)$ and $S_B = S(B;\gamma)$.

Label	Operational definition	Illustrative pair
Equivalent	$S_A = S_B$, including applicable scope and exceptions.	Inclusion age ≥ 18 versus age ≥ 216 months after a valid age-unit conversion.
A more restrictive	$S_A \subsetneq S_B$.	A: inclusion age ≥ 21; B: age ≥ 18.
B more restrictive	$S_B \subsetneq S_A$.	The preceding pair reversed.
Partial overlap	Nonempty intersection, with neither set containing the other.	Inclusion ages 18–65 versus 50–80, all endpoints inclusive.
Contradictory or disjoint	$S_A \cap S_B = \varnothing$ in the common operational frame.	Inclusion age ≥ 18 versus age < 18. This does not claim that the complete trials are mutually exclusive.
Related but not comparable	A meaningful clinical relationship exists, but a defensible shared constraint representation or mapping is unavailable.	Renal-function rules expressed through different measures without a supported conversion.
Unrelated	No meaningful direct clinical comparison for this task.	An age threshold versus an unrelated laboratory requirement.
Ambiguous	Missing or conflicting evidence prevents a defensible choice among the relationships.	A truncated rule whose polarity or governing exception cannot be established.

The distinction between "related but not comparable" and "ambiguous" is important. In the first case, the concepts are understood but the relationship cannot be ordered in the available representation. In the second, the meaning or relevant context of the input itself is insufficient. Annotators should record the reason, not force uncertain cases into an ordinary negative class. Empty or internally inconsistent allowed sets also require an explicit inconsistency disposition rather than a misleading containment label based on the mathematical empty-set convention.

3.4 Labeling decision procedure and worked safeguards

1. Confirm the source version and that both records are eligible for pair review. Inspect parent statements and any linked atoms needed for scope.
2. Identify the clinical quantities, anatomical/clinical context, applicable population, measurement method, temporal window, and exceptions.
3. Classify clear unrelatedness, related non-comparability, or unresolved ambiguity. Otherwise establish the common comparison frame.
4. Translate each rule into its criterion-level allowed predicate. Validate units/references, boundary inclusivity, internal negation, and eligibility role.
5. Determine equality, proper containment, intersection, or separation. Record a concise evidence-supported explanation and any assumptions.
6. Reverse A and B as a consistency check. Only the two restrictive labels swap; symmetric labels remain unchanged.

For example, inclusion "age ≥ 18" and exclusion "age ≥ 18" are not equivalent: the latter allows ages below 18 under the simplified common age universe. Similarly, changing a negation flag does not prove disjointness unless the complete predicates are complements. The allowed set for $\neg(\text{age} \geq 65)$ overlaps the set for age ≥ 18. Different lookback windows can change the relationship even when entity, threshold, and units match. A single atom from $A \wedge B$ is not an independent exclusion unless its conjunction condition remains explicit.

3.5 Annotation record and review protocol

Annotation component	Required content
Identity	Ordered pair ID, unordered pair-group ID, corpus version, A/B criterion IDs and trial IDs.
Evidence	Source excerpts, parent/context references, source versions/links, and field-level differences used by the reviewer.
Sampling	Every candidate-generation method that selected the pair, retrieval model/version and rank/score, phenomenon tags, and natural/constructed flag.
Decision	Eight-class label, direction, rationale, assumptions, source-ambiguity reason, reviewer uncertainty, and review status.
Adjudication	Reviewer identity, initial independent decisions when available, final adjudicator, revision history, and unresolved disposition.
Evaluation control	Family/group IDs, split, annotation-guide version, duplicate/variant links, and whether the case was used for development or demonstration.

JM owns pair definitions, construction, labeling, adjudication, and evaluation changes. JG consumes the approved development records for software and model work and does not redefine gold labels to fit outputs. Independent second annotation of a documented subset is desirable; if a second reviewer is available, preserve the initial disagreement and JM's adjudication rather than reporting only the consensus. The guide should be piloted and revised before the held-out benchmark is locked. Model suggestions and similarity scores should be hidden during an independent gold-labeling round where feasible, reducing anchoring on the method being evaluated.

3.6 Family splits and pair sampling

Map trials to families before candidate generation. Related protocols, amendments, template-derived duplicates, and constructed variants must stay in one partition. A 70/15/15 family allocation is a starting configuration, not a guarantee of balanced criteria or relationship classes. Audit the actual family counts, oncology/cardiovascular composition, domains, and pair opportunities; record a fixed seed and the complete mapping. Missing split assignments must cause a pair-generation failure or explicit rejection, not silent inclusion in the known side's partition.

Candidate pools should combine canonical-entity matches, nearest neighbors from multiple encoders, lexical neighbors, random controls, and targeted clinical phenomena. Record overlapping sampling memberships instead of letting whichever generator ran first erase the others. Natural pairs measure behavior in the observed corpus; constructed paraphrases and controlled threshold/negation changes test diagnostic capabilities and are reported separately. Random controls mainly provide easy unrelated cases and must not dominate a benchmark intended to test subtle clinical differences.

4 Agentic Architecture

4.1 Runtime workflow and tool contracts

The system has an offline data/model workflow and an online comparison workflow. Offline work freezes the corpus and splits, builds candidate pools, collects labels, trains models when justified, selects policies on development data, and prepares versioned indexes. Online inference uses frozen weights and versions; reviewer corrections enter an audit log and a later offline release rather than silently changing a running model.

The input contract contains the two records, necessary parent evidence, corpus/split identifiers, requested comparison mode, and a tool budget. A tool output should contain a supported label or unresolved result, explicit preconditions, field differences, cited evidence, assumptions, and a requires-context flag. Scores must identify their

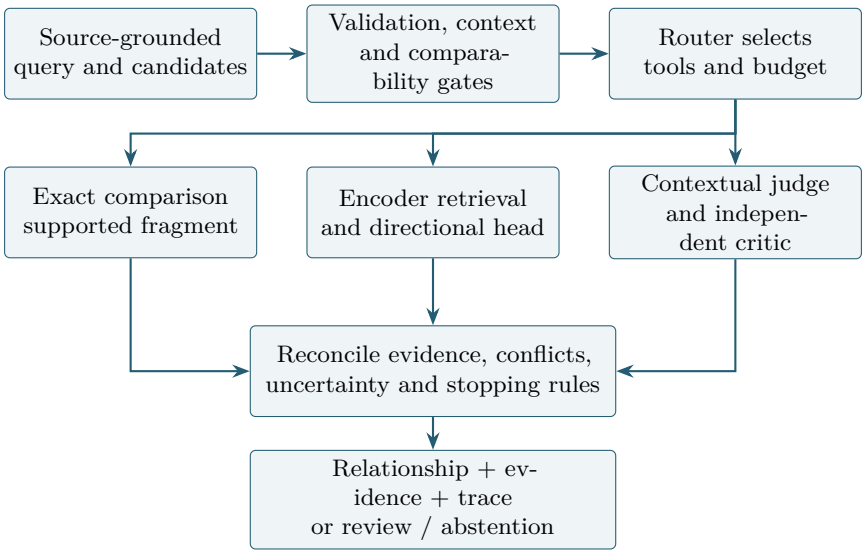


Figure 1: Proposed runtime. Retrieval proposes comparisons; routing decides which additional reasoning is required. A route can use several tools without making every request execute the same fixed chain.

meaning: cosine similarity, uncalibrated classifier score, calibrated probability, or a deterministic logical result are different objects.

The initial router can be a transparent rule policy. Numerical records with matching supported quantity/context signatures may enter the exact route; large candidate searches use the encoder; exceptions, ambiguous history, measurement incompatibility, and unresolved scope invoke contextual review. A learned routing policy is a later experiment requiring its own labels and cost/quality evaluation, not a prerequisite for an auditable prototype.

4.2 Exact Comparisons

4.2.1 Supported fragment and admission gates

The exact route is a conservative comparator for an explicitly supported fragment: scalar thresholds, complete intervals, supported unit conversions, categorical predicates with controlled semantics, and simple temporal windows with shared anchors. More complex temporal expressions, clinical hierarchies, assay conversions, and nested Boolean conditions require additional representation or contextual reasoning.

Before emitting an exact relationship, check entity identity, measurement method, applicable population, units/reference basis, temporal anchor, repetition, exceptions, negation, and Boolean scope. Numerical validity alone does not satisfy these gates. Missing context is not a wildcard match. An unsupported case returns a reason for escalation rather than a fabricated exact answer.

4.2.2 Interval algebra and directional containment

Represent a numerical constraint by a lower endpoint, lower inclusivity, upper endpoint, and upper inclusivity. Missing endpoints denote unbounded directions only where the comparator actually establishes that meaning. For example, $x > t$ becomes $(t, +\infty)$ and $x \leq t$ becomes $(-\infty, t]$. Strict and inclusive boundaries remain distinct.

Compare the resulting allowed sets, including any complement introduced by an exclusion rule. First check equality, then proper containment, then disjointness, and finally genuine partial overlap. At a shared endpoint, intersection exists only if both intervals include that point. Excluding an interval can yield a union of two intervals; a comparator that supports only one interval must escalate that case or implement a verified interval-union representation. It must not drop one side of the complement.

An exact explanation should expose the transformation, for example: "A allows $[21, +\infty)$ and B allows $[18, +\infty)$; A is a proper subset of B." A witness such as age 19 can illustrate why B allows an additional case. Such a witness

is a logical example, not an actual patient or an estimate of how many people a trial would enroll.

4.2.3 Units, reference ranges, and time

Use versioned, deterministic conversion rules only for commensurable quantities. Record both the source and converted value. An absolute concentration cannot be compared to a ULN multiple without a shared reference value and assay context. Drug washouts expressed in half-lives similarly require a drug-specific half-life and an approved interpretation before a fixed-day conversion is possible.

Temporal comparison needs the event, anchor, direction, duration, and inclusivity. "Within 30 days before first dose" differs from "30 days after the last dose." A rule using the lesser of 28 days or five half-lives is a conditional duration expression, not a single fixed bound. Endpoint tolerances and investigator exceptions should remain explicit. Repetition requirements need a compatible observation definition and spacing rule; an extra observation is not automatically a stricter criterion if its sampling or allowed timing differs.

4.2.4 Boolean scope, negation, and exceptions

Do not assume a flat list of atoms is a complete formula. In an exclusion $(A \wedge B) \vee C$, the record for A must remain conditional on B, and C remains an alternative branch. The v1 parent/context fields preserve evidence, but exact Boolean execution should require an explicit, reviewed expression tree or an equally precise scoped representation. If this is unavailable, use contextual review. An SMT solver is an optional future extension, not evidence that the current flat schema already supports arbitrary logic.

An exception is part of the governing rule. If a disqualifying malignancy predicate is M and an allowed cured-cancer exception is E, the exclusion predicate can be $M \wedge \neg E$ when that accurately reflects the source. Encoding E as an independent exclusion reverses its role. Internal negation, categorical absence, and exclusion type must be reconciled once, explicitly.

4.2.5 Current implementation boundary

The repository contains a starter interval comparator and deterministic-first orchestrator. They are scaffolding for this specification. The current comparator does not implement proper-subset direction, complete inclusion/exclusion set semantics, or all of the context/reference gates above. Its hard-coded confidence values are not calibrated probabilities. The exact-route acceptance suite should therefore include boundary, containment, complement, scope, unit, reference, and temporal cases before the route is described as clinically reliable.

4.3 The Encoder Model

4.3.1 Role, input representations, and inference

The encoder supports scalable candidate retrieval and a representation for supervised pair prediction. Given serialized criterion text $t(c)$, produce a unit vector

$$z_c = \frac{f_\theta(t(c))}{\|f_\theta(t(c))\|_2}, \qquad s(A, B) = z_A^\top z_B.$$

The text serializer is part of the experiment configuration. An initial fair comparison supplies identical raw criterion text to all baselines. Additional ablations can use a structured rendering or a source-plus-context rendering, but their results should be reported as different input conditions. Preserve comparators, units, negation, timing, reference basis, eligibility role, and necessary qualifiers in a structured rendering. Never include gold labels, reviewer rationales that reveal them, or held-out annotation metadata.

The planned general baseline is `sentence-transformers/all-mpnet-base-v2`; its model card describes a 768-dimensional sentence representation and default truncation beyond 384 word pieces [4]. The biomedical baseline is `FremyCompany/BioLORD-2023`, which builds on MPNet with definition/ontology-grounded biomedical training [5]. For each model, pin the revision and tokenizer, inspect truncation, use the specified pooling implementation, and save normalization and input settings. Do not silently omit a critical exception at the end of a long parent.

Retrieval excludes the query itself and, for the cross-trial task, its trial. Every experiment declares its permitted candidate partition. Cache keys include corpus hash, serializer version, model revision, and inference configuration. A cached embedding matrix is invalid when any of these changes.

4.3.2 Directional relationship head

A shared encoder with cosine similarity cannot distinguish A-more-restrictive from B-more-restrictive. A proposed ordered head uses

$$h_{AB} = [z_A; z_B; z_A - z_B; |z_A - z_B|; z_A \odot z_B; \phi(A,B)], \qquad p(y \mid A, B) = \mathrm{softmax}(g_\psi(h_{AB})),$$

where ϕ contains approved structured differences and missingness flags. This is a proposed feature design, not a selected final architecture. Compare a simple frozen-embedding classifier against encoder fine-tuning before adding complexity. Structural features must be derived from the input, not gold adjudications. A high classifier score does not authorize ignoring failed comparability gates.

4.3.3 Supervision and training objectives

Training data may contain equivalences, ordinary negatives, hard negatives, and ordered anchors. Hard negatives include lexical neighbors whose polarity, boundary, timing, or exception changes the relationship. They should not include unlabeled pairs arbitrarily declared negative. Restrictive and overlapping pairs remain conceptually related; treating every non-equivalent pair as a maximally distant negative would damage useful retrieval geometry.

One training option is a supervised contrastive loss on justified positives, an ordered classification loss, and a diagnostic ordering loss:

$$\mathcal{L} = \lambda_{\mathrm{con}}\mathcal{L}_{\mathrm{con}} + \lambda_{\mathrm{cls}}\mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{ord}} \max\{0, m - s(a,p) + s(a,n)\} + \lambda_{\mathrm{swap}}\mathcal{L}_{\mathrm{swap}}.$$

Here (a, p, n) is an explicitly reviewed relative-similarity triple, not an ordering inferred from restrictiveness alone. The swap term penalizes a failure to reverse directional labels while preserving symmetric ones. Loss weights, margin, negative sampling, early stopping, and class weighting are selected on development data. Report learning curves and at least three training seeds for the final trained comparison. If supervision is inadequate or no improvement is observed, retain the strongest frozen model and report the result.

4.3.4 Retraining procedure and release checklist

Proposed design. Retraining is an offline maintenance operation, never part of a demonstration comparison. This checkout has encoder inference wrappers but no implemented training entry point. The following is the required runbook for that implementation; it does not describe an executable training command that already exists. The training deliverable must add exact preparation, training, evaluation, export and index-building commands to the README, together with a checked-in example configuration.

1. **Decide what needs updating.** Record the reason: new adjudicated pairs, a changed labeling contract, reproducible errors, or a new base model. Adding corpus criteria alone requires encoding/index refresh, not necessarily encoder retraining. Changing deterministic rules alone requires route validation. A label-contract change requires a versioned annotation migration and reevaluation of affected examples before training.

2. **Freeze the inputs.** Select an immutable corpus release and manifest, adjudicated ordered-pair labels, label mapping, serializer and family/group split manifest. Keep existing test families held out; group duplicates, templates and swapped pairs together. New examples enter the appropriate partition without moving old test cases into training. Save input hashes and validate IDs, label direction, missingness and leakage.

3. **Recreate the environment and configuration.** Save the code revision, dependency lock, hardware, base-model ID and full resolved commit. Specify frozen-head training versus encoder fine-tuning, feature schema, maximum sequence length, truncation audit, loss functions and weights, negative/triple sampling, class weights, optimizer, learning rate, schedule, batch size, gradient accumulation, epochs or step limit, seeds, checkpoint frequency, stopping criterion and development selection metric. Record all effective values, including defaults. Resolve choices on development data.

4. **Estimate resources before the full run.** Run a short training smoke run and measure peak memory, step time and validation overhead on the chosen machine. A frozen-embedding head is the lower-cost first experiment; full encoder fine-tuning may benefit from a separate GPU machine. Reduce batch size or use validated gradient accumulation if memory is insufficient; record the effective batch and any change to negative sampling. Do not claim identical contrastive behavior merely because effective batch sizes match.

5. **Train and select without test feedback.** Generate only authorized training examples, fit learned preprocessing on training data, and save loss curves, development metrics and checkpoints. Select using the predeclared development policy. Save optimizer/scheduler state, random states and data-order configuration for resumable checkpoints; distinguish resuming an interrupted run from starting a new experiment. Repeat the final selected configuration with at least three seeds and retain failures as well as gains.

6. **Evaluate before promotion.** Compare the candidate with the current release and frozen baselines on identical partitions. Report retrieval and ordered-label metrics, false equivalence, swap consistency, clinical error slices, abstention/coverage, and local runtime and memory. Tune calibration or abstention only on development data. Run held-out test evaluation after configuration selection; do not repeatedly tune against it. If the candidate fails the declared quality or resource gates, retain the previous release and report the negative result.

7. **Export and rebuild compatible assets.** Export encoder, tokenizer, pooling/normalization configuration, relationship head, label map and serializer. Verify that reloaded outputs match the selected checkpoint within recorded tolerances. Rebuild corpus embeddings whenever encoder, serializer or corpus changes; a head-only change can reuse unchanged embeddings but still requires a new compatible application manifest.

8. **Release and rehearse.** Publish the approved artifacts and model card to the Hub, pin the resulting full commit, and retain the previous release for rollback. Package evaluation reports, configuration, seeds, hashes and index manifest. Download the release to the demonstration machine and verify offline startup, representative comparisons, latency and replay before selecting it as the default. Publication credentials stay outside the configuration and model package.

Training duration is not yet measured. Estimate it from the smoke run's steps per epoch, planned epochs and validation cost; report actual elapsed time and peak memory with the resulting release. Model training and demo inference may use different machines without requiring a hosted website.

4.3.5 Retrieval versus interpretation

A useful retrieval candidate can be an equivalent rule, a stricter rule, or a clinically important contradiction. Define the relevance policy for each retrieval experiment rather than assuming that only equivalent pairs matter. Training can pursue broad concept retrieval while the ordered head and exact tools distinguish clinical relationships. Evaluate both stages so a retrieval gain does not conceal a rise in false equivalence.

4.3.6 Hugging Face model release and local inference

Proposed design. Publish the trained TriAlign encoder as a versioned Hugging Face Model Hub repository. The Hub stores and distributes model checkpoints [2]; the demonstration downloads the selected release in advance and executes inference on the presenter's computer. The model repository identifier will be assigned when a trained release is ready. Publication is a release task, not a claim that a TriAlign checkpoint already exists.

The model release should contain encoder weights, tokenizer files, pooling and embedding-normalization settings, and the configuration needed to load the encoder. Package the ordered relationship head, label-to-index mapping, and approved structured-feature serializer alongside it when those components are trained. If the head has a separate artifact, pin both artifacts together in the application release manifest. The model card should identify the base checkpoint and its resolved revision, training-corpus and split versions, training configuration, evaluation results and limitations, and applicable redistribution terms. Keep held-out labels and local API credentials out of the model package.

Pin the full resolved Hub commit for each encoder and relationship-head release. A moving branch name alone is insufficient to reproduce a run. Hugging Face supports revision-specific downloads and local caching [3]. During

preparation, download all required model files, verify the selected revision and checksums, and run an inference warm-up. During the demonstration, load the prepared local model directory rather than resolving a newer remote revision. A missing or mismatched artifact should produce an actionable startup error instead of a silent download or model substitution.

Precompute corpus embeddings with the same model revision and serializer. At runtime, encode a new query locally, retrieve from the local index, and run the relationship head and exact tools on the same machine. CPU execution is the initial deployment target; locally available acceleration can be used after compatibility and latency are measured. Switching encoder releases requires regenerating the corpus index and validating its manifest.

This architecture supports model distribution through Hugging Face while keeping application execution local. A public website, a Hugging Face Space, and a dedicated inference endpoint are not required for the planned demonstrations. A hosted application can be evaluated as a later deployment option after the research workflow and operating requirements are established.

4.4 Fuzzy Clinical Reasoning: LLM Judge and Critic

4.4.1 Tasks and evidence-bounded prompts

The contextual route handles cases where an exact interpretation is not licensed by the supported fragment: implicit clinical language, exceptions, severity/history distinctions, incompatible measurements, treatment settings, or unresolved branch scope. The judge receives the ordered pair, source excerpts, necessary parent/context, reviewed structured fields, and any validated tool evidence. Retrieved neighboring criteria can provide precedent only when the experimental policy allows them; they cannot establish facts missing from the two source rules.

The prompt requests a structured answer: proposed label, direction, cited evidence spans, field/context differences, assumptions, uncertainty reason, and review/abstention disposition. Instructions distinguish absent evidence from evidence of absence. The model should name a missing temporal anchor or assay mapping rather than supply a clinically plausible one from memory. Source text is untrusted task data: embedded instructions in a protocol or retrieved document cannot authorize tool calls or override the task policy.

4.4.2 Independent criticism and bounded revision

The critic is a separate invocation that checks the proposed answer against the original evidence and any exact-tool certificate. Its checklist covers source support, inclusion/exclusion interpretation, negation, units and references, interval direction, timing, exceptions, branch scope, and unsupported assumptions. It also checks whether the label remains correct when the input order is reversed.

The critic may accept the supported result, identify specific repairable errors, or require abstention/reviewer escalation. One bounded revision followed by a final evidence check is an initial policy to evaluate; repeated self-discussion is not a substitute for missing evidence. A second call to the same model can have correlated errors, so critic agreement is not an independent human gold label or proof of correctness. Measure the critic's incremental benefit with a no-critic ablation.

4.4.3 Validation, reconciliation, and uncertainty

Validate schema, allowed label values, valid criterion IDs, required evidence, and quote grounding before accepting a model response. Invalid JSON, missing decisions, corrupted text, unsupported evidence, timeouts, and refusal/error responses are operational failures with explicit dispositions. They must not silently become unrelated or equivalent labels.

If the judge conflicts with the exact route, first recheck the exact route's preconditions. A mathematical result is conclusive only for its encoded predicate; the encoding itself may be wrong or incomplete. An unresolved conflict triggers escalation. Do not reconcile incompatible labels by taking an unvalidated weighted average of cosine, a classifier score, and model self-reported confidence.

Separate *source ambiguity* from *model uncertainty*. The former concerns what can be established from the inputs; the latter concerns the system's reliability in choosing an answer. Confidence calibration requires held-out

development labels. Until then, expose raw scores and categorical reasons for review. Select any automatic-accept threshold on development risk–coverage results, then freeze it for test evaluation.

4.5 Trace, stopping policy, and operational behavior

Each comparison receives a trace with ordered input IDs, corpus/source hashes, serializer and normalization versions, tool/model/prompt versions, route reasons, evidence, outputs, conflicts, final disposition, and human edits. Record wall-clock latency, token usage, configured pricing date when cost is reported, cache hits, retries, and tool errors. Store task evidence and auditable summaries, not a claim to expose private internal model reasoning.

The prototype should have explicit budgets for candidate count, tool calls, revision attempts, timeout, and cost. A valid exact result can stop without an LLM call; unresolved language can stop with abstention when the evidence is insufficient. Training and benchmarking must record these policies so reduced cost cannot be explained by silently dropping hard cases.

5 UI/UX Design

5.1 Clinical users and decisions to support

Design status: exploratory. The interface should be shaped by task observation and feedback from clinicians and trial staff. The following personas are design hypotheses rather than findings from a completed user study.

- A protocol investigator asks which historical rules operationalize the same concept and how a proposed change affects restrictiveness.
- A research coordinator needs a source-grounded explanation of a flagged difference and a way to send it for investigator review.
- A reviewer or methodologist needs to compare multiple protocols, record adjudications, and export an evidence trail for a design or harmonization meeting.

The primary interface question remains the proposal's practical task: given a current rule, which historical criteria are comparable, how do they differ, and what evidence supports the result? Avoid presenting the tool as a patient-screening interface when it has neither patient inputs nor validated screening logic.

5.2 Core interaction and information architecture

Proposed design. A three-region workspace can put a query and source context on the left, a ranked candidate list in the center, and a selected comparison on the right or in an expanded lower panel. A compact view should answer the clinical question before showing implementation traces.

1. Select a source trial and criterion, or enter a draft rule. If draft text is parsed, display the parsed interpretation for confirmation and mark it as an unreviewed draft rather than part of the frozen corpus.
2. Inspect the source wording, parent, role, population, time, and provenance.
3. Retrieve candidates with filters for therapeutic area, clinical concept, trial, source version, and comparison status. Clearly show which filters constrain the search; exclusion of a candidate is not a clinical decision.
4. Select a candidate and inspect its relationship, the decisive differences, supporting source spans, and any unresolved assumptions.
5. Accept, correct, or leave unresolved with a rationale. Distinguish a human review action from a system's automatic-accept recommendation.
6. Export the reviewed comparison with source and version identifiers.

A search result should show the candidate's source excerpt, trial, clinical concept, and relevance score type. A relationship label appears only when comparison has actually run. "No results" should distinguish restrictive

filters, no candidate match, unavailable indexes, and tool failure. A blank or failed route must not look like a confident negative finding.

5.3 Visualization ideas retained from the proposal

These ideas remain candidates for the design, with implementation priority based on clinical usefulness and feasibility.

Concept	Useful presentation	Initial priority / limitation
Side-by-side constraint comparison	Align entity, role, comparator/value, units/reference, interval, timing, repetition, qualifiers, Boolean scope, and source; annotate the decisive difference.	Highest priority. Keep source language accessible beside normalized fields.
Numerical allowed-region plot	Draw the two admissible sets with open/closed endpoints, intersection, and containment; label the common quantity and units.	High for supported numerical cases. Draw allowed regions after polarity handling; hide the plot when context/conversion gates fail.
Agent decision trace	A compact route diagram expands to tool inputs, evidence, conflicts, critic disposition, and final action.	High. Show the real executed trace, not a decorative flowchart implying uncalled tools ran.
Method-agreement panel	Compare lexical/encoder rankings, exact result, judge, critic, and final decision without collapsing their distinct meanings.	High. Label cosine as similarity, not probability.
Uncertainty and review indicator	Show supported result, needs review, or unable to determine, with a specific reason and calibrated probability only when available.	Medium. Source ambiguity and model confidence are different dimensions.
Similarity versus clinical relationship plot	A cohort-level plot identifies false-high and false-low similarity by reviewed label and clinical phenomenon.	Medium, after labels exist. Appropriate for results/error analysis rather than the primary comparison screen.
Two-dimensional retrieval map	An optional projected neighborhood colored by reviewed relationship and marked by route.	Exploratory. Projection distances are not literal clinical distances and can distort the original geometry.

5.4 Additional clinician-oriented interface ideas

5.4.1 Protocol comparison matrix and review packet

A clinician may start with two protocols rather than one criterion. A comparison matrix could list clinical concepts as rows and protocols as columns, highlighting matched rules, missing matches, and unresolved scope. An unmatched rule means no supported match was found; it does not prove that the other full protocol lacks that requirement. A review packet could export selected comparisons, excerpts, source versions, reviewer rationale, and open questions for a protocol-design meeting.

5.4.2 Amendment difference view

For two versions of one protocol, separate wording-only changes from changed thresholds, windows, exceptions, or measurement methods. Show a proposed direction of change only for supported comparable rules. Preserve both versions and the amendment lineage. This is a useful future workflow, but amendments must remain in the same family for evaluation.

5.4.3 Source-first evidence drawer and context expansion

Every result can open an evidence drawer with highlighted source spans, parent statement, adjacent context, snapshot version, and registry link. Link a clinical summary such as "longer treatment washout" to the exact two windows that support it. Offer an explicit "context not sufficient" disposition instead of making the reviewer search for a hidden caveat.

5.4.4 What-if comparison sandbox

A draft workspace could let a user change one threshold or window and inspect the logical relationship to the original. Keep draft versions separate from the frozen corpus, and require confirmation of a parsed rule. A numerical witness can illustrate the set difference. Do not translate that difference into recruitment percentages or clinical safety claims without the additional data and validation those claims require.

5.4.5 Disagreement and adjudication queue

A review queue can prioritize exact/LLM conflicts, missing evidence, uncertain scope, or method disagreement. Show why each case was queued and support assigning a reviewer, adding an interpretation, and preserving earlier decisions. An override requires a rationale and evidence; a corrected result does not erase the model's original prediction from the evaluation record.

5.4.6 Saved cohorts of comparisons and precedents

Users may collect a set of source-backed precedents for a clinical concept, bookmark unresolved questions, and compare the same draft against several historical protocols. Group results by distinct trial/family so repeated template wording does not create a false impression of broad independent precedent. Saved sets should record filters and source versions for replay.

5.5 Accessibility, uncertainty, and usability validation

Use text and icons in addition to color; show full labels rather than only red/green agreement. Support readable typography, keyboard navigation, sortable lists, and expandable evidence without forcing a dense field table on every user. Prefer plain clinical explanations with technical details available on demand. Distinguish "evidence not available", "not comparable", "system error", and "review pending".

Evaluate prototype tasks with trial staff: find a supported precedent, identify a consequential threshold or scope difference, reject an unsupported equivalence, and complete an adjudication/export. Record task success, time, corrections, overlooked caveats, and users' interpretation of score labels. This is a proposed usability study; no clinician preference or efficiency gain is assumed. A within-person comparison with and without the tool should counterbalance task order if used to measure time savings.

5.6 Demonstration scope and resilience

For November 17, target the ranked list, aligned constraint view, supported interval plot, genuine route trace, and a method-agreement view. Add a confidence indicator only if its interpretation is defensible. By December 8, include the error-analysis plot and a documented uncertainty/review policy. The projected map and larger protocol matrix are optional extensions.

The presentation application starts on the presenter's computer and is displayed in a local browser. This is the planned demonstration deployment for both November 17 and December 8, avoiding recurring website and managed inference hosting costs. Model distribution follows Section 4.3.6; the runtime uses locally prepared weights, the frozen corpus, and a precomputed embedding index.

The live demonstration must include a success, an important failure, and a necessary abstention. Live local execution means the selected local tools actually run. Calling the external contextual LLM remains an optional online operation with its own API usage cost and network dependency. A local-only mode runs retrieval and

supported exact/model comparisons, and explicitly escalates a case if it needs unavailable contextual reasoning. A cached mode replays saved versioned outputs, including saved judge/critic results where available, and is labeled as replay. A new input without a matching cached case cannot receive a fabricated replay result.

A recording or screenshots provide a final fallback. Before presentation, verify local startup, model/index revisions and checksums, inference warm-up, expected latency on the demonstration machine, source evidence, local review storage, and replay mode. Check credentials and connectivity only for the optional online route; stored source excerpts remain available locally even when registry links cannot be opened. The current Streamlit application is a candidate-table placeholder, not an implemented version of this workspace.

5.7 Local startup and demonstration assets

The existing local UI entry point can be launched from the repository root with the project's Python environment and application dependencies installed:

```
python -m streamlit run app/review_app.py \
  --server.address 127.0.0.1 --server.port 8501
```

Open http://127.0.0.1:8501 in the presenter's browser. This command currently starts the placeholder review surface; completing the agent workflow behind that entry point remains an implementation deliverable. The intended finished workflow should use this documented local launch path, with any additional preparation/configuration steps listed in the README. Binding to the loopback address supports a single-machine demonstration without a public deployment.

Local asset	Preparation and runtime requirement
Python environment	Install and pin dependencies before the presentation; record the application Git revision and machine configuration.
Model files	Download the pinned Hub release in advance; retain weights, tokenizer, configuration, relationship head and checksums on disk.
Corpus and search index	Load the frozen Parquet/manifest and matching precomputed embeddings/index. Check corpus hash, model revision and serializer version together.
Demonstration cases	Save selected inputs, source excerpts, expected routes, output versions and representative success/failure/abstention cases.
Replay bundle	Keep complete saved outputs and traces locally; require matching input and configuration identifiers before replay.
Review records	Persist corrections, rationale and traces to a local SQLite database or versioned local files; preserve them across application restarts.
Optional LLM route	Read credentials from the local environment, enable online calls explicitly, and set call/time/cost limits. Without that route, use local-only execution or clearly labeled replay.

The startup workflow should distinguish a valid warm cache from missing or incompatible assets. Rehearse from a fresh application process after all downloads are complete, including an offline run. The offline acceptance criterion is that the UI starts, local retrieval and supported comparisons work, local evidence is readable, and prepared cases can be replayed without network access. Arbitrary new fuzzy comparisons still require a running contextual model or escalation. Local startup removes the need for website hosting; it does not imply that training, initial model downloads, or optional external LLM inference have no resource or usage costs.

5.8 Compute requirements and demonstration latency

Proposed design. The initial goal is an interactive single-user comparison on the presenter's machine. No end-to-end latency measurements exist yet. A trained encoder performs inference rather than retraining for each input.

With a loaded model and precomputed corpus embeddings, a new query needs one encoding, local retrieval, and comparisons for a bounded candidate set. Comparing two new criteria needs their encodings; comparing stored criteria can reuse compatible cached vectors. Deterministic checks do not generate text and are expected to contribute less delay than contextual LLM reasoning. These are architectural expectations, not measured performance claims.

For the current 3,537-criterion corpus, a 768-dimensional float32 embedding matrix occupies approximately 10.4 MiB before metadata/index overhead. An exhaustive query scan evaluates about 2.7 million vector components, which makes simple local vector search a reasonable initial design. Model weights, activations, tokenization and application overhead consume additional memory. Do not recompute the whole corpus or enumerate all criterion pairs in response to a single UI action. A full protocol comparison is a different workload: batch its query encodings and bound the candidates and contextual calls per criterion.

The main latency risks are cold model loading, long inputs, repeated encodings, and the contextual judge/critic route. An external LLM introduces network and service latency; sequential judge and critic calls add to the critical path. A local generative LLM may instead be limited by RAM, available accelerator memory and token generation speed, and can compete with the encoder for resources. Running the encoder locally does not establish that the complete generative route will be fast or fit on the same laptop. Backend, device and batch-size optimizations must be benchmarked on the actual machine rather than assumed to improve performance [1].

Keep the encoder and index loaded across UI interactions, warm them before presenting, reuse compatible embeddings and bound the retrieved top-k. Invoke contextual reasoning only when the route requires it, with explicit call, output-token, timeout and revision limits. Show retrieved evidence and completed exact results while contextual work is pending. On timeout, retain those results and display an unresolved state; offer a matching, explicitly labeled replay when available. Do not silently skip a required critic or present an incomplete relationship as final to meet a speed target.

Benchmark and acceptance procedure. Record machine model, CPU, RAM, accelerator, software/model/index revisions, input lengths and top-k. Measure cold startup separately from warm comparisons. Use representative exact, encoder-only and judge-plus-critic cases, including long inputs and failures; report sample counts, median and 95th-percentile latency, peak memory, cache status, route/call counts and external API time. Measure stage times for encoding, search, exact/head inference, judge, critic and rendering as well as the user-visible total. A proposed usability target is a warm local retrieval/exact result within 2 seconds and a completed contextual comparison within 30 seconds. These are targets to validate and revise, not promised runtimes. If the contextual route misses the budget, preserve the fast local workflow and use a budgeted external model, suitable local acceleration, or clearly labeled prepared replay for the demonstration.

6 Evaluation and Comparisons

6.1 Benchmark construction and evaluation units

Maintain separate pair-classification and retrieval evaluations. Pair classification tests a fixed set of ordered comparisons with reviewed labels. Retrieval tests a query and a declared pool of candidate criteria. A model can classify supplied pairs correctly while failing to retrieve them, or retrieve relevant concepts while confusing their relationship. Report both stages.

Use family splits before mining, preserve reversed/constructed variants in their source partition, and keep test labels unavailable to training, prompt design, model selection, agent memory, or retrieval of labeled precedents. Encoding held-out candidate text for a declared test search index is different from training on it; document the intended inductive evaluation. Fit TF–IDF vocabulary/IDF and all learned preprocessing on training text only. Select hyperparameters, calibration, and acceptance thresholds on development data. Run the untouched test evaluation after those choices are frozen.

Pool candidate judgments across the proposed baselines so one encoder does not determine the entire benchmark. Report judgment coverage, label counts, sampling strata, and the number of queries with known relevant candidates. Unjudged items are not automatically negatives. If the candidate pool is incompletely judged, call recall relative to that judged pool and disclose that it is not exhaustive corpus-wide recall.

6.2 Baseline Embedding Comparisons

Baseline	Fixed implementation / role	Key controls
Lexical	TF–IDF with lowercase word unigrams/bigrams and cosine ranking.	Fit only on training data; retain operators/negation in an audited serializer and record tokenization limitations.
General encoder	Frozen `sentence-transformers/all-mpnet-base-v2`.	Pin revision, pooling, truncation and input rendering.
Biomedical encoder	Frozen `FremyCompany/BioLORD-2023`.	Same candidate pool and input condition; document model-card license requirements before redistribution.
CReSE-informed representation	Eligibility-oriented representation/rephrasing approach, conditional on reproducible artifacts.	Distinguish an actual reproduced checkpoint from a newly implemented inspired baseline.
TriAlign encoder	Fine-tuned representation plus an ordered prediction head, when sufficient labels exist.	Same held-out evaluation; compare with frozen features and report training seeds.

The raw-text baseline is the simplest matched input condition. A second condition may provide the same structured/context rendering to every method. Do not compare a context-rich TriAlign input against context-deprived baselines without separately identifying that input advantage. A separate relationship classifier can be trained on each frozen embedding under the same protocol to isolate representation quality from the value of fine-tuning.

The model inventory in configuration also lists MiniLM, BGE, and SapBERT. They are optional candidates, not mandatory experiments or established improvements. Add one only with a documented question, pinned revision, supported input policy, and sufficient time to run the same evaluation. The required initial comparison remains TF–IDF, MPNet, and BioLORD.

6.3 Existing Eligibility Systems and Products: Research Notes

The following entries are research notes for the systems named in the proposal. Most are research methods or software artifacts, not interchangeable commercial clinical products. No shared eight-class benchmark results or TriAlign adapters are claimed here. Availability of a paper or repository does not establish installation success, compatible licenses, usable model weights, or reproduction of the published results.

6.3.1 CReSE: eligibility recommendation and representation

Research note; adaptation not yet implemented or evaluated. The published work introduces eligibility-criterion recommendation from trial information and an evaluation framework using contrastive learning and LLM rephrasing [6]. Its representation objective makes it a relevant comparator for eligibility-specific embeddings; its native task is not TriAlign's directional eight-label comparison.

Proposed comparison. If a suitable checkpoint or reproducible training path is available, evaluate criterion embeddings on the same retrieval pools, with any relationship head trained only on TriAlign training pairs. Otherwise report a clearly named CReSE-inspired experiment rather than claiming to reproduce CReSE.

Open research items. Identify code/checkpoints, exact preprocessing, data access and license, and possible source-trial overlap. Determine whether rephrasing preserves strict comparators, exclusion roles, and exceptions. Exit artifact: an availability/reproduction note and, only if feasible, a matched retrieval result.

6.3.2 TrialGPT: patient-to-trial matching

Research note; adaptation not yet implemented or evaluated. The official repository describes retrieval, criterion-level patient eligibility matching, and trial ranking, and provides code and data [8]. These inputs and outputs differ from comparing two protocol criteria. Its original patient-matching labels cannot be relabeled as TriAlign relationships without a justified task transformation.

Proposed comparison. Inspect its evidence-and-judgment prompt design and implement a separately named criterion-pair adaptation if defensible. Hold the supplied source evidence, base LLM, and allowed call budget constant when isolating prompt/workflow differences. Compare evidence support, abstention, latency, and cost on shared cases; preserve native results as background rather than importing their accuracy into a TriAlign leaderboard.

Open research items. Pin a repository revision, identify reusable prompts and output schemas, document all changes, and determine whether a pair adaptation is substantive enough to warrant inclusion. The cited clinical study remains background for the original task [7].

6.3.3 EligMeta: eligibility-aware evidence synthesis

Research note; adaptation not yet implemented or evaluated. EligMeta connects trial discovery and eligibility-aware meta-analysis. The paper separates LLM-generated rules/parsing from deterministic logic, weighting, and statistical pooling [9]. Its relevant overlap is eligibility alignment and reproducible execution; TriAlign does not implement its pooled-outcome estimand.

Proposed comparison. Investigate whether an exposed alignment component can operate on the common source-grounded cases. If not, compare architectural choices and a documented adaptation of its rule/parse/execute pattern. Never compare TriAlign pair macro-F1 directly to meta-analysis effect estimates or literature-selection recovery.

Open research items. Verify code and artifact access, alignment granularity, treatment of exceptions and absent fields, and compatibility of its similarity output with categorical relationships. Determine a fair matched-case input/output contract before implementing an adapter.

6.3.4 SatIR: constraint-satisfaction retrieval

Research note; adaptation not yet implemented or evaluated. SatIR formulates trial matching using explicit SMT constraints and scalable relational-algebra retrieval. The inspected paper is arXiv version 4 and links an implementation [10, 12]. Its native problem is profile-to-trial retrieval rather than ordered criterion-pair labeling.

Proposed comparison. Study whether its constraint representation can support a restricted, reviewed pair-comparison fragment. Equality, containment, and separation can be investigated through satisfiability of set-difference/intersection formulas, provided both predicates and a common universe are faithfully encoded. A solver's answer does not validate the LLM translation that produced the formula.

Open research items. Record ontology assumptions, missing-data policies, supported temporal/Boolean expressions, source translation errors, and installation dependencies. Report unsupported coverage together with correctness on supported cases. Keep retrieval-system reproduction separate from a newly written logical pair adapter.

6.3.5 VERDICT: accountable decisions and rationales

Research note; adaptation not yet implemented or evaluated. VERDICT translates constraints and decision policies into SMT and uses SMT/MaxSMT reasoning to support explicit assumptions and pivotal conditions. It evaluates groundedness, consistency, and counterfactual self-faithfulness in clinical-trial matching [11]. The inspected paper links code/prompts in the Stanford OVAL repository [12].

Proposed comparison. Adapt its rationale-oriented questions: which source facts support the relationship, which assumptions are required, and what supported change would alter the decision? Test whether a stated pivotal condition actually changes a recomputed result. Treat these as additional audits of TriAlign explanations, not automatic proof that an explanation is clinically correct.

Open research items. Identify reusable policy/solver interfaces, the boundary between facts and assumptions, coverage of conditional exceptions, and the cost of faithful translation. A common-case adapter must declare every modification to its native decision task.

6.3.6 EC-RAFT: retrieval-augmented eligibility generation

Research note; adaptation not yet implemented or evaluated. EC-RAFT generates eligibility criteria from trial titles and descriptions using retrieval-augmented fine-tuning. Its paper identifies Llama-3.1-8B-Instruct as the base model and links released code [13]. Generation quality is a different task from pair relationship prediction.

Proposed use. Treat EC-RAFT as adjacent authoring research. It may inform a future draft-criterion workflow or supply separately marked constructed examples after source/clinical review. Generated criteria are not automatically valid training labels or clinical precedents.

Open research items. Check the generation/evaluation interface, retrieval-corpus overlap, and preservation of numerical and exception scope. Report any generation experiment separately from the main retrieval and relationship benchmark. It should not displace the mandatory matched-task baselines.

6.3.7 Common research-note template and comparison decision

For each system, maintain primary citation, inspected version/date, native task, input/output schema, artifact/license status, reproducible execution status, proposed adapter, task mismatch, expected cost, and comparison decision. Classify the outcome as: direct matched-task baseline; explicitly adapted common-case experiment; architectural/failure-mode comparison; or out of scope. Commercial products can be added later with the same template if the team identifies a specific accessible product and evaluation pathway. Do not invent a capability matrix from marketing claims or compare unavailable systems using fabricated results.

6.4 Agent comparisons and ablations

The main agent experiment compares an LLM-only judge, an always-run fixed chain, and routed TriAlign on the same supplied pairs, evidence, model versions, and output contract. Include an exact-only route with supported-case coverage where useful. Report both accuracy and expenditure; an expensive always-run system can improve accuracy while being less efficient.

Ablations isolate hard negatives, human corrections, normalization/context input, structured features/checks, critic calls, and targeted sampling. Change one component at a time where possible and state interactions when several components change. Routing evaluation should include selection errors, missed opportunities to stop exactly, unnecessary LLM calls, and unresolved conflicts. Do not score an abstaining exact-only system solely on its easy resolved subset and compare that number to full-coverage models.

6.5 Metrics, uncertainty, and statistical reporting

Evaluation target	Measures	Interpretation controls
Retrieval	Precision/recall@k, MRR, equivalence-specific retrieval, reviewed anchor ordering.	State relevance definition, candidate/judgment pool, query denominator and judgment coverage.
Pair relationships	Eight-class macro-F1, per-class precision/recall, confusion matrix, directional F1 and swap consistency.	Keep class imbalance and asymmetric labels visible; report abstentions separately.
Clinical phenomena	Error counts/rates for thresholds, negation, units/references, timing, repetition, history, exceptions and compounds.	Give numerator/denominator and account for multi-tag cases.

Evaluation target	Measures	Interpretation controls
Uncertainty	Calibration diagrams, Brier score where probabilities are defined, risk–coverage, inappropriate automatic acceptance, escalation/abstention errors.	Calibrate on development data; model self-report is not validated probability.
Evidence and explanation	Supported-claim rate, missing-condition errors, source-span support, contradiction and pivotal-condition checks.	Correct label and grounded rationale are distinct outcomes.
Workflow	Reviewer time and corrections, end-to-end latency, tool/LLM calls, tokens, cost, cache hits, failures.	Use matched tasks and comparable budgets; show live and cached execution separately.

At automatic-accept threshold t, define coverage as the fraction of all evaluated cases accepted automatically and selective risk as the error rate among those accepted. Report how many ambiguous cases are appropriately escalated and how many answerable cases are unnecessarily deferred. Uncertainty estimates should preserve trial-family dependence. Use paired method comparisons and family-aware resampling; a cross-trial pair touches two families, so an ordinary independent-pair bootstrap is not sufficient. One option is to resample held-out families and retain/weight pairs through their endpoint-family memberships, applying the same resample to all methods. Predeclare the chosen scheme and summarize sensitivity rather than implying that one simple bootstrap removes every dependency.

Report at least three seeds for trained models, effect sizes and intervals, sample sizes, family counts, and the limits of a 100-trial convenience cohort. Separate natural from constructed challenges. The final report must show representative false-high/false-low similarity, a routing or interpretation failure, and a necessary abstention alongside successes.

A Implementation, Delivery Milestones, and Reproducibility

A.1 Repository boundary and work ownership

The repository currently provides criterion/pair schemas, workbook and Parquet adapters, candidate generation, family-splitting utilities, a TF–IDF runner, a lazy Sentence Transformers wrapper, a starter exact comparator, a deterministic-first orchestrator, and a placeholder review application. These interfaces do not establish that fine-tuning, contextual judging, calibration, full family-aware command-line splitting, or the proposed UI have been implemented. The pair generator's unordered ID and the split/comparator integration must be reconciled with this specification before benchmark production.

Workstream	Jonathan Ma (JM)	Jiatong Gao (JG)
Data and benchmark	Source release, normalization, family mapping, pair guide, construction, labels, adjudication, source validation and manifests.	Consume frozen development inputs and report interface/data issues without rewriting gold labels.
Models and agent	Define comparison semantics, metrics, and case-level evaluation requirements.	Implement and run TF–IDF/MPNet/BioLORD, routing/tools/critic, traces, comparison adapters where feasible, and UI-facing outputs.
Evaluation	Freeze splits/labels, audit directionality and reported case evidence.	Save predictions, rankings, runtime/cost logs, ablations and model/configuration artifacts.

Workstream	Jonathan Ma (JM)	Jiatong Gao (JG)
Submission	Data/evaluation documentation, test-case index, provenance and contribution record.	Baseline/agent implementation documentation, reproducible execution, prompts, traces and UI assets.

Shared work includes reviewing changes, agreeing on contracts, inspecting successes and failures, rehearsing demonstrations, integrating the report, and recording contributions by artifacts and commits. JM retains authority over gold labels and evaluation-set changes; JG owns baseline and orchestrator implementation. Proposed independent second annotation should be arranged without confusing an annotator's input with final adjudication authority.

A.2 Milestone 1: October 20 presentation, completed

The first presentation introduced the clinical problem, source-data pipeline, 100-trial cohort, structured comparison, embeddings, agent design, and initial review work. Recorded feedback requested eligibility-focused system comparisons and consideration of sharing a documented criterion subset with another clinical-trials group working with JEV. Any such sharing preserves public source provenance, intended use, and evaluation-label separation.

A.3 Milestone 2: November 17 presentation

The format is 15 minutes plus 5 minutes of Q&A. The presentation must cover objective, dataset, workflow, preliminary positive and negative results, a live UI, limitations, and next steps.

1. JM delivers the frozen criterion release and data dictionary, approves the pair-labeling contract, establishes family splits before pairing, and provides a versioned pilot with evidence and directional consistency checks.
2. JG runs the three required frozen baselines on common inputs, prepares TrialGPT/EligMeta capability and adaptation notes, implements an auditable prototype router, and compares it with LLM-only/fixed-chain execution on the supplied development pairs.
3. Together, integrate ranked candidates, source/context, aligned constraints, supported numerical plots, real route traces, and cached replay. Save at least one success and one meaningful failure; demonstrate abstention when needed. Launch the application locally using Section 5.7, with all required model and index assets prepared in advance. Verify startup and replay without an internet connection.

The v1 criterion freeze is complete. Pair labels and implementation/evaluation artifacts remain subsequent deliverables. An approximately 200–300-pair pilot is an optional planning target, not an existing benchmark or a guaranteed class-balanced sample. If fine-tuning is not ready, report frozen baselines rather than an unvalidated training run.

A.4 Milestone 3: December 8 final presentation

The format is 20 minutes total for presentation, demonstration, and Q&A. The final package for this milestone includes the frozen corpus/labels and family splits; reproducible baseline and agent configurations; trained-model results if justified; comparative evaluation, ablations and uncertainty analysis; documented positive/negative cases; and stable local live/replay demos.

JM finalizes and validates the benchmark and all denominators. JG completes baseline runs, agent routing/reconciliation/critic/escalation, feasible external-system adapters, and UI integration. Freeze dataset, model, prompt, normalization, routing and calibration versions before the untouched test run. Do not change gold labels to accommodate method errors; genuine source corrections require a transparent versioned amendment.

A.5 Milestone 4: December 22 report and submission

The report is 5–8 pages excluding references and permitted appendices. This technical document is supporting material and is not constrained to that report length. The report should cover the clinical objective; source/final data and normalization; AI/agent workflow; benchmark and statistical methods; results and important failures; limitations and future improvements; and individual contributions.

The accompanying release should contain source data or a reconstruction manifest; criterion/pair manifests and family/split files; labeling guide and adjudication provenance; code and prompts; model/index/configuration versions; per-case inputs, predictions, evidence, traces and review outcomes; summary tables generated from saved outputs; and reproducible training instructions or model artifacts. Include the Hub repository identifiers and full resolved release revisions, a local model/index manifest, hardware and latency notes, and instructions for preparing an offline demonstration bundle. The README must cover installation, dependencies, API access for the optional online route, local startup, expected outputs, replay, reproduction and troubleshooting.

Verify a clean-environment execution of data loading, baseline inference, agent comparison, evaluation and local demonstration. Rehearse local-only, optional online LLM, and cached replay modes separately, including restart persistence of reviewer decisions. Link every report example to its frozen case and source evidence. Record cache behavior and the limits of reproducing stochastic external-model outputs. Freeze a tagged submission archive after the contribution statements and numerical claims are audited.

A.6 Risks, mitigations, and definition of done

Risk	Mitigation and reporting requirement
Insufficient pair labels	Prefer a smaller well-supported benchmark, preserve uncertainty, seek independent annotation of a subset, and report learning curves.
Extraction/normalization errors	Preserve raw evidence and correction lineage; version changes; escalate unsupported reference/context assumptions.
Leakage	Map families before mining; group amendments/templates/variants; restrict fitting, prompt development and labeled retrieval memory to approved partitions.
No fine-tuning or routing gain	Report negative findings and ablations; retain the strongest justified frozen configuration.
Unfair external comparisons	Require a matched output contract or identify the work as an adaptation/architectural comparison; do not combine incomparable native metrics.
Uncalibrated confidence	Label scores accurately; select thresholds on development data; report coverage and error together.
Live/API failure	Start locally with predownloaded models and indexes; use bounded retries for optional online calls, explicit error states, cached replay, and a rehearsed recording fallback.
Deployment cost	Run the demonstration on the presenter's machine and distribute versioned model artifacts through the Hub. Track any training or optional LLM API costs separately from website hosting.
Clinical overinterpretation	Keep criterion-level inference, logical witnesses, and protocol precedent distinct from patient screening, population prevalence and causal conclusions.

Done means another person can reconstruct the approved data release, run the documented workflow, reproduce principal results within stated tolerances, inspect every evaluation case and source, operate the demonstration, and distinguish source evidence, model proposals, human adjudication, and unresolved outcomes. Completing the interface or producing a high aggregate score alone does not satisfy that definition.

References

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[5] FremyCompany / F. Remy et al. *BioLORD-2023 model card*. https://huggingface.co/FremyCompany/BioLORD-2023.

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[8] NCBI NLP. *TrialGPT: official code and data*. https://github.com/ncbi-nlp/TrialGPT.

[9] Y. Zhao, Z. Zhang, and Y. Xu. *Eligibility-Aware Evidence Synthesis: An Agentic Framework for Clinical Trial Meta-Analysis*. arXiv:2604.02678v1, 2026. https://arxiv.org/abs/2604.02678v1.

[10] Z. Zhou et al. *SatIR: Scalable High-Recall Constraint-Satisfaction-Based Information Retrieval for Clinical Trials Matching*. arXiv:2604.08849v4, 2026. https://arxiv.org/abs/2604.08849v4.

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[13] N. Lekuthai et al. *EC-RAFT: Automated Generation of Clinical Trial Eligibility Criteria through Retrieval-Augmented Fine-Tuning*. Findings of ACL, 2025. https://aclanthology.org/2025.findings-acl.491/.

[14] F. Kury et al. *Chia, a large annotated corpus of clinical trial eligibility criteria*. Scientific Data 7, 281, 2020. https://www.nature.com/articles/s41597-020-00620-0.